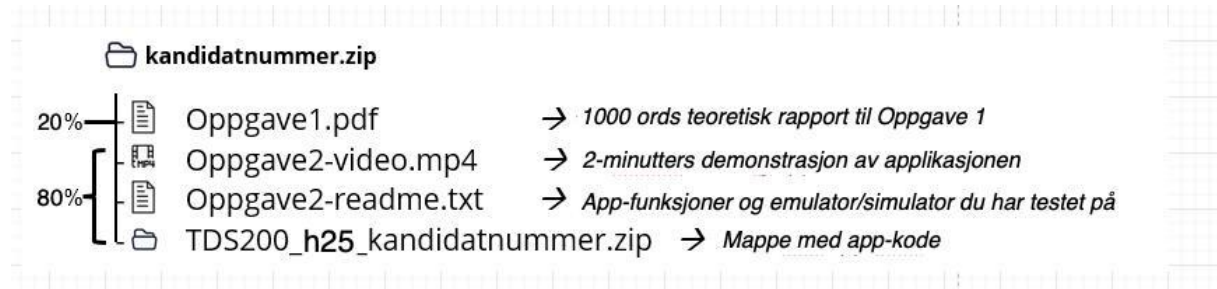

You will be required to answer two tasks, and your performance will be assessed on both technical and theoretical competence. The following pages contain detailed descriptions of the two tasks. The submission should follow the structure outlined below.



- **Plagiarism should not occur. Copying code without clearly citing the source is not permitted.**
- **Use of component libraries and npm modules is not considered plagiarism, and you are encouraged to demonstrate your expertise by using such third-party libraries. Remember to cite sources here as well.**

Task 1 – Theoretical competence (weighting: approx. 20%)

In this assignment, you will read through the following letter from Nordic Digital Solutions.

Topic: Starting the DugnadHub app

Dear Development Team,

Following our collaboration with the Office of Municipal Affairs, we are now starting up The DugnadHub project—a digital platform that will strengthen local engagement and simplify the organization of volunteering in the local community. The application will modernize the Norwegian volunteering tradition by bringing together citizens, volunteers and organizers in one intuitive and collaborative interface.

Our team consists of two full-stack developers with expertise in JavaScript/TypeScript and is responsible for delivering a cross-platform solution for both web and mobile devices. The platform will enable local communities to publish volunteer events, describe tasks, manage sign-ups, and facilitate real-time communication through messaging and photo sharing.

To ensure accessibility and relevance, we will conduct market research on mobile usage in Norway (Android vs. iOS), which will form the basis for design and development. The development will emphasize efficiency, security and scalability, and utilize cloud infrastructure to achieve both short-term delivery goals and long-term sustainability.

As demand for digital solutions for local community engagement increases, time to market is a critical factor. Our strategy must balance rapid development with long-term maintainability and cost-

effectiveness. VolunteerHub will contribute to increased participation, shared responsibility and local pride – in line with the municipality's goal of strengthening cooperation in the local community through technology.

**With kind regards,
Nordic Digital Solutions**

Based on the information above and the requirements specification below, you, in your role as Tech Lead, will design and justify a recommendation on the most appropriate approach to the development of VolunteerHub app, and present this in a professional report of approximately 1,000 words with at least 3-5 relevant external sources, such as professional literature, research articles, online material, professional blogs or video content. The reference list should not be counted in. The report should also contain illustrations and flowcharts that support the analysis.

#	Function
1	The app should communicate with a backend service, such as Firebase.
2	Users should be able to register and log in with email/password. Authenticated users should be able to volunteer.
3	Authenticated users should be able to see upcoming volunteer opportunities in a list and open the details page by clicking on an entry.
4	Authenticated users should be able to sign up for a volunteer position.
5	Organizers should be able to create, edit, and manage volunteer events in a secure and intuitive way. New event entries can include a title, description, required tasks, location, date/time, number of volunteers needed, and contact information.
6	Each volunteer post should show remaining volunteer spots (for example, 3 out of 10 volunteers are needed).
7	Each event detail page should include photos and a list of attendees, and may include progress updates.
8	Volunteers should be able to leave comments, share updates, and show recognition (for example, with likes or badges) to promote engagement and collaboration.
9	Users should be able to see their own statistics, including participation in past, ongoing and upcoming volunteer efforts.
10	Seamless navigation between pages/views is expected.
11	The app should have a clean and functional visual design.
12	The app will be available on both Android and iOS to cover the entire mobile market.

Task 2 – Technical competence (weighting: approx. 70%)

In this assignment you will develop the application VolunteerHub.

Main criteria for assessment:

- The solution should be able to be built and run without errors, and be able to run in a browser as well as on an Android emulator and/or iOS simulator.
- Comment the code so that understanding is clear, and explain complex logic where necessary.
- The code should follow good practices for quality: appropriate names, clear structure, and emphasis on reusability.
- The application should be user-friendly, and all functionality should work as expected.
- Any external libraries, modules or packages that the project requires must be integrated correctly and appropriately.

Creation and delivery of exam project:

- Create your own Firebase project, and make sure there is data there, including at least one test user (provided in a separate .txt file) and some sample tasks, so that the tester can log in and verify that the application is working as expected.
- Create a short video (screen recording without audio) that demonstrates the features you have implemented in your application. The video should not exceed 2 minutes and should show any advanced features that may affect the rating.
- Attach a file named Task2-readme.txt which briefly describes what functionality you have implemented in the app, and whether you have tested it in the Android emulator, iOS simulator, or both.
- Remove node_modules folder before uploading-
- Make sure that package.json contains all the necessary npm dependencies, as your team leader will install the packages with the command npm install.
- The project folder you submit should be named TDS200_h25_your candidate number.zip.
- The delivery must contain a .env-file with the necessary Google API keys if you have used relevant Google services. These keys are project-specific, and you can safely delete the associated Google Cloud project after the exam has been graded.

Case: DugnadHub app

This technical assignment builds on the theoretical analysis in Assignment 1. Regardless of the conclusion you drew in Assignment 1 about the recommended development approach, the DugnadHub application will be developed with React Native for the Frontend and Firebase as the Backend-as-a-Service, in line with the technologies reviewed in the course.

Four main requirements for the implementation (numbering without priority):

- **User registration:** Users can register and log in with email/password to participate as volunteers.
- **Viewing of volunteer work:** Registered users can view upcoming volunteer opportunities in a list. Each entry can be opened for a detailed view showing description, category, photos, location, number of volunteers needed, etc. Registered users can sign up for volunteer opportunities.
- **Creation of Dugnades:** Registered users can create new volunteer jobs by specifying a title, description, required tasks, images, number of volunteers, and date/time. The solution should support images via both camera and file upload from image gallery.
- **Seamless navigation:** Users should be able to navigate smoothly between different pages/views.

The four points above constitute the minimum requirements for the assignment. To achieve a higher grade, more functionality and greater complexity in the implementation are expected. See the examples below for features that can contribute to a higher grade. Note that the app should be designed so that the team leader can easily test the solution with minimal manual configuration.

Examples of extended functionality that can contribute to a higher grade:

Each function is marked with a difficulty level in parentheses (Easy, Intermediate or Advanced) to provide guidance on expected complexity and workload.

- **(Simple) Search and filter function:** Users can search and/or filter volunteer opportunities based on keywords such as location, category, etc.
- **(Simple) Error handling:** Implement user-friendly mechanisms for handling errors and alerts.
- **(Simple) Favorites list:** Users can mark efforts as favorites.
- **(Medium) Personal statistics:** Users can view their own volunteer data via the profile page, including participation in past, ongoing, and upcoming volunteer activities, and a favorites list.
- **(Medium) Interactive communication:** Users can rate events, leave comments, share updates, and show recognition (such as with likes or badges) to drive engagement.
- **(Medium) Event Management:** Each volunteer position shows total capacity and remaining volunteer slots (for example, 7 out of 10 slots filled). Available slots are dynamically updated in real time as users sign up or withdraw.
- **(Medium) Geocoding:** Using geocoding to convert addresses to coordinates.
- **(Advanced) Extended Authentication:** The application supports multiple login methods, including email/password and social logins like Google Sign-In.
- **(Advanced) User Roles:** Distinguish between organizers and volunteers. Organizers can create, edit, and delete events, as well as view and manage participant lists. Volunteers can view, sign up for, and withdraw from volunteer activities.

- **(Advanced) Admin dashboard:** A separate page to monitor statistics, users and participations (for organizers or administrator only).
- (Variable level) Other functional or non-functional improvements that demonstrate independence, creativity, or technical skill.

- Some key learning objectives
- Navigation
- User authentication
- Protected routes
- Native plugins (camera, geolocation, google maps, etc)
- Local storage
- State
- Callback functions
- Modal
- Flatlist
- Navigation with expo-router
- Native APIs
- Permissions
- Camera
- Photo album

- Teknologier
 - VS Code (eller hva du liker best)
 - Typescript
 - Expo
 - React Native
 - Tailwind CSS

